

Boba “ Feature Status (all components)

Revision: 10 **Last modified:** 2026-08-21T00:00:00Z **Scope:** Every system component, service, infrastructure piece, and client app of the Boba project “ one row per REAL unit (endpoint / handler / client method / component control / plugin / subcommand / script), grouped by component. **Authority:** assembled by READ-ONLY repo inventory (codegraph + grep + source reading) on 2026-06-15, expanded to per-unit granularity (Â§11.4.118 discovery-pressure) on 2026-06-16. Cross-references AGENTS.md, CLAUDE.md, docs/REMAINING_WORK_PLAN.md.

Rev 9 (2026-08-20, BOB-083 /

GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-16)

“ staleness catch-up (Â§11.4.44/Â§11.4.45/Â§11.4.86): 161 commits landed between the Rev 8 pass (e6162f7, 2026-08-18) and this pass’s HEAD (e0d60ab, 2026-08-20). Real git log e6162f7..HEAD evidence: the overwhelming majority are the BOB-126 forced-logout killpg(-1, SIGKILL) incident chain (root cause found + closed, new universal Â§11.4.263 anchor) plus surrounding test/gate/quality hardening (BOB-127..135) “ process-safety and CI-discipline work, **not new product features**, so no new feature rows are warranted from that set (Â§11.4.6: verified by reading the commit log, not assumed). Two exceptions **were** feature-relevant and are corrected below with cited live evidence: **RW-06** (RuTracker ReDoS fix deployment) “ CLOSED live 2026-08-19 (1c0389a, BOB-093); **RW-07** (nnmclub /auth/nnmclub/status SOURCE“ARTIFACT drift) “ CLOSED live 2026-08-19 (7baef2b, BOB-092). Both rows below updated to PASS “ LIVE with their real evidence paths. docs/REMAINING_WORK_PLAN.md itself was NOT edited (out of this task’s owned-file scope “ see the Status_Summary companion for the plain-language version of this note). Total features cataloged: **unchanged at 292** (no rows added or removed this pass, only 2 existing rows’s Validation cells corrected). Every other pre-existing row was left untouched “ a full per-unit re-audit against the 161-commit delta was judged out of the bounded scope of this staleness-catch-up pass, per the same Â§11.4.6 option-B discipline Rev 8 used.

Rev 8 (2026-08-18, BOB-075) “ staleness remediation (Â§11.4.44 / Â§11.4.86 /

GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-08):

the audit’s staleness claim was CONFIRMED, not a false positive “ a real content edit landed in ea86ce1 (2026-08-15, the docs_chain-engine row swap in Â§8d below) WITHOUT the Revision/Last-modified header being bumped, and 60 commits landed between the Rev 7 content commit (646b295, 2026-06-16) and this pass (64705eb, 2026-08-18) that were never reflected at

row-granularity. This pass performed a TARGETED refresh (Â§11.4.6 option B â€” this document is narrative, â€œassembled by READ-ONLY repo inventory,â€” not mechanically derivable from a single source-of-truth table, so a full per-unit re-audit of all 289 pre-existing rows was judged out of the bounded scope of a staleness-fix task and is NOT claimed here): added 3 new rows to Â§8d for scripts that shipped since Rev 7 and were completely absent from this ledger â€” scripts/load-tracker-cookies.sh (tracker-cookies autoloader, 619a5f6), scripts/boba-svc.sh (sudo-free systemd wrapper, 444d94e), scripts/commit-push-all.sh (Â§11.4.234 dedicated commit/push endpoint, 08e870e) â€” each cited to its real source file + covering challenge. Total features cataloged: 289 â†’ **292**. Every pre-existing row was left untouched (no content was invented to â€œlook freshâ€”); the honest scope boundary of this pass is recorded under â€œComponents NOT inventoriedâ€” below.

Rev 7 (2026-06-16) â€” live-PASS update for the search/auth fix batch (HEAD 7e9cab5): the multi-word URL-encoding fix (da7d709, ~17 nova3 plugins), the new RUTRACKER_COOKIES injection (2fc29fc), NNMCLUB_COOKIES auth, /auth/status cookie reflection (9c2f8dc), the new GET /api/v1/healthz JSON endpoint (137d7ff), and Â§11.4.85 stress+chaos coverage for the multi-word fix (7e9cab5) were all PROVEN end-to-end on the live nezha stack â€” full-fleet the matrix = **2600 results / 23 contributing trackers / zero plugin_bad_query_encoding**, all four private trackers authenticate (rutracker 50, nnmclub 50, kinozal 50, iptorrents 49). Evidence: docs/qa/search-fix-verify-20260616/. kickass.py reclassified **Wonâ€™t-fix structurally-impossible (Â§11.4.112)** per docs/research/kickass_403_20260616/. NOTE (Â§11.4.6): no Go files were touched this session â€” the Go rows are unchanged (the pre-existing magnet_stress_chaos_test.go guard stands; no NEW Go query-encoding guard was added this session, so none is claimed here).

Captured-evidence-driven (Â§11.4.5 / Â§11.4.45 / Â§11.4.86). Every feature row cites a real source file (file:line where load-bearing), endpoint, command, or control. **No invented features** (Â§11.4.6). As of Rev 5 the **Video** column is DEFINITIVE for every row â€” there is no bare PENDING. Each cell is exactly one of: VIDEO-CONFIRMED â€” <file> (the feature is shown on-screen in a committed recording), PENDING (UI â€” film next) (a user-visible control/dialog not yet *individually* filmed), or N/A (no UI â€” test-covered + exercised by <journey>) (a non-user-visible unit â€” REST endpoint / handler / client method / parser / scanner / service / crypto / repo / script â€” which has no screen of its own and is confirmed by its cited tests plus the UI/CLI journey that drives it). **Honest classification (Â§11.4.6/Â§11.4.143): a row is VIDEO-CONFIRMED only when a real recording actually shows it.**

Per-row Video tally (291 feature rows â€” the pre-existing Rev 8 tally of 288 vs the stated 289â€” total was a 1-row pre-existing discrepancy not reconciled by this pass, 11.4.6 honest note; +3 for the Rev 8 additions, all N/A/host-script): **28 VIDEO-CONFIRMED, 14 PENDING (UI “ film next), 249 N/A (no UI “ test-covered).**

Column legend: - **Impl** = implemented / partial / stub / missing - **Wired** = reachable on a running default-profile stack (yes / no / opt-in / host) - **Tests** = test types present + cite file (unit / integration / e2e / security / stress-chaos / property / contract); none if no covering test found - **Validation** = honest validation posture (tested-green-in-suite / unit-only / live-evidence-captured / not-validated / operator-blocked) - **Video** = DEFINITIVE video-recording status “ one of VIDEO-CONFIRMED “ <file> / PENDING (UI “ film next) / N/A (no UI “ test-covered + exercised by <journey>)

Video-recording confirmations (11.4.143 / 11.4.83)

Real-use recordings (Claude-vision analyzed “ the HelixAgent ensemble vision is a proven stub, see docs/qa/recording-readiness-20260615/) in / Volumes/T7/Downloads/Recordings/ (project-prefix boba-). Per-row Video cells stay PENDING where a feature has not been *individually* filmed; the recordings below confirm the headline user-facing flows on-screen.

Recording	Surface	Features confirmed	Verdict
boba-cli-orchestrator-demo.mp4	CLI (boba-ctl)	status / health / list “ 4 services running & healthy launch “ search “debian“ “ live results “ “829 results (288 merged)“ complete dashboard load (29 trackers, qBit connected), result rows render qBit + Download	docs/qa/recordings-20260615, boba-cli-verdict.md
boba-web-search-flow.mp4	Web dashboard	“live results “ “829 results (288 merged)“ complete dashboard load (29 trackers, qBit connected), result rows render qBit + Download	boba-web-search-flow-verdict.md
boba-web-feature-tour.mp4 + boba-web-dashboard-tour.mp4	Web dashboard	“live results “ “829 results (288 merged)“ complete dashboard load (29 trackers, qBit connected), result rows render qBit + Download	boba-web-feature-tour-verdict.md

Recording	Surface	Features confirmed	Verdict
boba-extension-{scan-detect,popup,options,tour}.mp4	BobaLink extension (MV3, real --load-extension Chromium)	buttons (clickable), tab nav (Results/Active Downloads/Trackers/Schedules/Hooks), Jackett /jackett credentials page, theme content-script detects a magnet + appends a ðŸŒŒ MAGNET badge (non-torrent correctly skipped); popup UI (header, server status, Refresh + Send All, Detected-torrents); options page (7 settings tabs + server config) Active Downloads / Schedules / Hooks tabs each render their empty-state + columns; theme toggle; processing spinner (â€œFound N resultsâ€¦!â€¦); bridge/qBit-Connected header indicator; Jackett Add-credential	boba-extension-verdict.md
boba-web-tabs-theme-tour.mp4 (+ frames boba-web-tab-active-downloads.png / boba-web-tab-schedules.png / boba-web-tab-hooks.png / boba-web-theme-toggle.png) + boba-web-jackett-add-credential.png	Web dashboard + Jackett page		Claude-vision verified 2026-06-16

Recording	Surface	Features confirmed dialog (Name/ Username/ Password/ Cookies + Cancel/Save)	Verdict
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Honest scope (11.4.6, no bluff): - **Backend services** (download-proxy / merge / boba-jackett / Go) verified **200 server-side inside the VM** and exercised through the web UI recordings. - **Console errors** seen in the web recordings are an **SSH-tunnel SSE/poll artifact** (macOS VM), NOT product defects – every endpoint is 200 in the VM. See the feature-tour verdict. - **BobaLink extension:** “... **VIDEO-CONFIRMED**” the prior --load-extension tooling gap was closed by building extension/scripts/record-features.mjs (loads the built MV3 artifact into a real Chromium, records the scan/popup/options journeys). Plus its passing test suite. - **TUI / mobile / desktop:** do not exist in this project (server-side + Angular web + browser extension only) – nothing to record.

Component summary

#	Component	Service / Port	Features cataloged
1	Download Proxy + Merge Search Service (Python/ FastAPI)	qbittorrent-proxy :7186 / :7187	69
2	qBitTorrent-go backend (Go/ Gin)	qbittorrent-proxy-go :7186/:7187/:7188 (opt- in --profile go)	47
3	boba-jackett (Go)	boba-jackett :7189	26
4	Tracker plugins (plugins/*.py)	run inside qbittorrent-proxy	30
5	Angular 21 frontend dashboard (frontend/)	served from :7187	34
6	BobaLink browser extension (extension/)	WXT MV3 client	39
7	WebUI bridge (webui- bridge.py)	host process :7188	4
8	Infrastructure / CLI / shell scripts	host	43

Total features cataloged: 292 (Rev 8: +3 “ scripts/load-tracker-cookies.sh, scripts/boba-svc.sh, scripts/commit-push-all.sh; Rev 7: +1 “ new GET /api/v1/healthz endpoint)

1. Download Proxy + Merge Search Service (Python/FastAPI) “ :7186 / :7187

Entry: download-proxy/src/main.py (starts legacy proxy thread + uvicorn api:app). This is the **shipped default product**.

1a. Merge Search Service “ REST API (download-proxy/src/api/routes.py)

Feature	Impl	Wired Tests	Validation
POST /api/v1/search “ async fan-out search, returns search_id (routes.py:283)	implemented yes	integration tests/integration/test_merge_api.py; e2e test_full_pipeline.py; stress test_merge_search_stress_chaos.py	tested-green-in suite + LIVE
POST /api/v1/search/sync “ synchronous search (routes.py:377)	implemented yes	integration test_merge_api.py; stress test_search_stress.py	tested-green-in suite; resets over tunnel (RW-08)
GET /api/v1/search/stream/{id} “ SSE result stream (routes.py:547)	implemented yes	integration test_realtime_streaming.py, test_streaming_browser.py	tested-green-in suite
GET /api/v1/search/{id} “ poll search state (routes.py:586)	implemented yes	integration test_merge_api.py	tested-green-in suite
POST /api/v1/search/{id}/abort “ cancel in-flight search (routes.py:619)	implemented yes	integration test_merge_api.py	tested-green-in suite
GET /api/v1/downloads/active “ qBit	implemented yes	integration test_buttons_api.py	tested-green-in suite

Feature	Impl	Wired Tests	Validation
active torrents (routes.py:629)			
POST /api/v1/auth/ qbittorrent â€” qBit WebUI login (routes.py:671)	implemented yes	integration test_login_actions.py	tested- green-in suite
POST /api/v1/download â€” add torrent to qBit (magnet/url) (routes.py:958)	implemented yes	integration test_button_functions.py, test_buttons_api.py; security test_ssrf_and_magnet_auth.py	tested- green-in suite; SSRF ga RW-03
POST /api/v1/download/upload â€” upload .torrent file to qBit (routes.py:1123)	implemented yes	unit tests/unit/api_layer/ test_download_upload.py	unit-only integrat
POST /api/v1/download/file â€” fetch tracker .torrent w/ auth cookies (routes.py:1212)	implemented yes	integration test_buttons_api.py; security test_ssrf_and_magnet_auth.py	tested- green-in suite; SSRF ga RW-03
POST /api/v1/magnet â€” resolve result â†’ magnet link (routes.py:1279)	implemented yes	integration test_magnet_dialog.py, test_buttons_api.py	tested- green-in suite; au gap RW-
GET /api/v1/theme â€” read theme state (routes.py:61)	implemented yes	contract test_crossapp_theme_contract.py	tested- green-in suite
PUT /api/v1/theme â€” persist theme (routes.py:67)	implemented yes	e2e test_theme_runtime.py, test_crossapp_theme.py	tested- green-in suite
GET /api/v1/theme/stream â€” SSE theme broadcast (routes.py:76)	implemented yes	e2e test_crossapp_theme.py	tested- green-in suite

1b. Auth API â€” per-tracker (download-proxy/src/api/auth.py)

Feature	Impl	Wired Tests	Validation
	implemented yes		

Feature	Impl	Wired Tests	Validation
GET /auth/ rutracker/ status "RuTracker session state (auth.py:56)		integration test_auth_and_links.py; e2e test_live_stack_evidence.py	partial " CA (BOB-008)
GET /auth/ rutracker/ captcha "fetch CAPTCHA image (auth.py:108)	implemented yes	integration test_auth_and_links.py	partial " op
POST /auth/ rutracker/ login "username/ password + CAPTCHA login (auth.py:214)	implemented yes	integration test_auth_and_links.py	partial " op
POST /auth/ rutracker/ cookie-login " cookie- based login (auth.py:298)	implemented yes	integration test_auth_and_links.py	PASS " LIV RUTRACKER_CO authenticates matrix = 50 r search-fix-v
GET /auth/ nnmclub/ status "NNM-Club session state (auth.py:350)	implemented yes	e2e test_live_stack_evidence.py:265 (no longer skips " see Rev 9)	PASS " LIV " RW-07 SO resolved: live returns HTTP JSON (docs/q nnmclub_stat the e2e"s S (7baef2b), tes recurrence ins
POST /auth/ nnmclub/ login "NNM-Club login (auth.py:407)	implemented yes	e2e test_live_stack_evidence.py	partial " live
GET /auth/ status "all-tracker aggregate	implemented yes	integration test_auth_state_ui.py; auth- status cookie-reflection test (1.1)	PASS " LIV reflects RUTRA NNMCLUB_COOK (9c2f8dc), so

Feature	Impl	Wired Tests		Validation
session state (auth.py:479)				rutracker/nnm fix-verify-2
POST /auth/ qbittorrent/ logout “ qBit WebUI logout (auth.py:532)	implemented	yes	integration test_login_actions.py	tested-green-i
Kinozal auth (status/login) REST route	missing	no	none “ no kinozal/login route in auth.py	not-implement via orchestrat
IPTorrents auth (status/ login) REST route	missing	no	none “ no iptorrents/login route in auth.py	not-implement via Jackett UI)

1c. Search orchestration internals (download-proxy/src/ merge_service/search.py)

Feature	Impl	Wired Tests	
Concurrency caps MAX_CONCURRENT_SEARCHES (429 on saturation)	implemented	yes	concurrency test_orchestrator_semaphore
Per-search tracker semaphore MAX_CONCURRENT_TRACKERS	implemented	yes	concurrency test_tracker_s
Public-tracker subprocess fan-out (NDJSON via patched novaprinter)	implemented	yes	e2e test_public_trackers_retu stress test_search_orchestration

Feature**Impl****Wired Tests**

Public-tracker deadline kill
PUBLIC_TRACKER_DEADLINE_SECONDS

implemented yes

chaos test_tracker_failure

Private-tracker aiohttp+cookie path
(rutracker/kinozal/nnmclub/
iptorrents)

implemented yes

integration test_tracker_auth_gated)

Per-tracker TrackerSearchStat
real-time stats

implemented yes

contract test_tracker_stats
property test_tracker_stats

Stderr error classification
(_classify_plugin_stderr)

implemented yes

chaos test_tracker_failure

TTL-bounded data stores
(cachetools) “no leak

implemented yes

memory
test_orchestrator_caches_

1d. Deduplication (download-proxy/src/merge_service/deduplicator.py)

Feature	Impl	Wired Tests	Validation	Video
Tier 1 dedup "metadata (imdb/tmdb) key"	implemented	yes property test_deduplicator_properties.py; benchmark test_deduplication_benchmark.py	tested-green-in-suite	N/A " - coverage exercised by w search " - merge results N/A " - coverage exercised by w search " - merge results
Tier 2 dedup "infohash key"	implemented	yes property test_deduplicator_properties.py	tested-green-in-suite	N/A " - coverage exercised by w search " - merge results N/A " - coverage exercised by w search " - merge results
Tier 3 dedup "name+size key"	implemented	yes property test_deduplicator_properties.py	tested-green-in-suite	N/A " - coverage exercised by w search " - merge results N/A " - coverage exercised by w search " - merge results
Tier 4 dedup "fuzzy Levenshtein name match"	implemented	yes property test_deduplicator_properties.py; benchmark test_deduplication_benchmark.py	tested-green-in-suite	N/A " - coverage exercised by w search " - merge results

1e. Enrichment " per metadata source (download-proxy/src/merge_service/enricher.py)

Feature	Impl	Wired Tests	Validation	Video
Enrichment: OMDb (enricher.py:113)	implemented	yes stress test_enricher_stress_chaos.py	tested-green-in-suite; needs OMDB_API_K	

Feature

Impl

Wired Tests

Validation

Enrichment:
TMDB

implemented yes

stress
test_enricher_stress_chaos.py

tested-green
needs
TMDB_API_K

Enrichment:
TVMaze

implemented yes

stress
test_enricher_stress_chaos.py

tested-green
in-suite

Enrichment:
AniList

implemented yes

stress
test_enricher_stress_chaos.py

tested-green
in-suite

Enrichment:
MusicBrainz

implemented yes

stress
test_enricher_stress_chaos.py

tested-green
in-suite

Feature	Impl	Wired Tests	Validation
Enrichment: OpenLibrary	implemented	yes stress test_enricher_stress_chaos.py	tested-green in-suite

1f. Tracker validation (download-proxy/src/merge_service/validator.py)

Feature	Impl	Wired Tests	Validation	Video
Tracker validation HTTP scrape BEP 48 (validator.py:101)	implemented	yes none dedicated found (covered via merge tests)	not-validated (no dedicated test)	N/A (no UI â€” exercised by web search flow seeds/leechers column)
Tracker validation UDP scrape BEP 15 (validator.py:162)	implemented	yes none dedicated found	not-validated (no dedicated test)	N/A (no UI â€” exercised by web search flow seeds/leechers column)

1g. Hooks (download-proxy/src/api/hooks.py + merge_service/hooks.py)

Feature	Impl	Wired Tests
GET /api/v1/hooks â€” list hooks (hooks.py:105)	implemented	yes unit test_hooks_coverage.py; concurrency

Feature	Impl	Wired Tests
		test_hooks_race.py; security test_hooks_schedules_auth.py
POST /api/v1/hooks " create hook (hooks.py:111)	implemented yes	unit test_hooks_coverage.py; concurrency test_hooks_race.py
DELETE /api/v1/hooks/{id} " delete hook (hooks.py:167)	implemented yes	unit test_hooks_coverage.py
GET /api/v1/hooks/logs " hook execution logs (hooks.py:179)	implemented yes	unit test_hooks_coverage.py
Hook event dispatch (8 events: search_start..validation_complete) (hooks.py:32)	implemented yes	concurrency test_hooks_race.py
Hook subprocess execution + sandbox (merge_service/hooks.py)	implemented yes	security test_no_shell_injection.py

1h. Scheduler (download-proxy/src/api/scheduler.py + merge_service/scheduler.py)

Feature	Impl	Wired Tests	Validation
GET /api/v1/schedules " list schedules (scheduler.py:39)	implemented yes	security test_hooks_schedules_auth.py	tested- green-in- suite; auth gap RW-02

Feature	Impl	Wired Tests	Validation
POST /api/v1/schedules "create schedule" (scheduler.py:64)	implemented yes	security test_hooks_schedules_auth.py	tested-green-in-suite; auth gap RW-02
GET /api/v1/schedules/{id} "get schedule" (scheduler.py:87)	implemented yes	security test_hooks_schedules_auth.py	tested-green-in-suite
PATCH /api/v1/schedules/{id} "update schedule" (scheduler.py:108)	implemented yes	security test_hooks_schedules_auth.py	tested-green-in-suite
DELETE /api/v1/schedules/{id} "delete schedule" (scheduler.py:126)	implemented yes	security test_hooks_schedules_auth.py	tested-green-in-suite
Scheduler engine (load/save/tick driver) (merge_service/scheduler.py)	implemented yes	integration test_dashboard_automation.py	tested-green-in-suite

1i. App-level routes / health / config / dashboard (download-proxy/src/api/__init__.py)

Feature	Impl	Wired Tests	Validation
GET /api/v1/healthz "machine-parsable JSON health probe" (routes.py:37)	implemented yes	integration test_merge_api.py	PASS "LIVE 2026-06-16" NEW endpoint (137d7ff): a bare /healthz was swallowed by the SPA catch-all; now mounted under /api/v1 returning JSON (11.4.43)
GET /health (__init__.py:168)	implemented yes	integration test_live_containers.py	tested-green-in-suite
GET /api/v1/bridge/health (__init__.py:173)	implemented yes	integration test_bridge_root_liveness.py	tested-green-in-suite
GET /api/v1/config (__init__.py:247)	implemented yes	integration test_merge_api.py	tested-green-in-suite
GET /api/v1/stats (__init__.py:276)	implemented yes	observability test_metrics_exist.py	tested-green-in-suite
	implemented yes	integration test_dashboard_rendering.py	

Feature	Impl	Wired Tests	Validation
Jinja2 dashboard GET / (__init__.py:337)			tested- green-in- suite
Jinja2 dashboard GET /dashboard (__init__.py:342)	implemented yes	integration test_dashboard_rendering.py	tested- green-in- suite
SPA catch-all GET /{path:path} (__init__.py:347)	implemented yes	integration test_dashboard_rendering.py	tested- green-in- suite

1j. Supporting subsystems (Python)

Feature	Impl	Wired Tests
Retry helpers (merge_service/ retry.py)	implemented yes	covered indirectly
Jackett autoconfig (Python, merge_service/ jackett_autoconfig.py) â€” legacy; canonical is boba-jackett	implemented yes	contract test_jackett_autoconfig_contract.p e2e test_jackett_autoconfig_e2e.py; benchmark test_jackett_autoconfig_perf.py

Feature	Impl	Wired Tests
Credential scrubbing log filter (config/log_filter.py)	implemented yes	security test_credential_scrubbing.py

1k. Legacy download proxy (plugins/download_proxy.py:7186)

Feature	Impl	Wired Tests	Validation
HTTP proxy â†’ qBittorrent WebUI w/ tracker URL interception (DownloadHandler:767)	implemented yes	integration test_bridge_header_rewrite.py	tested-green-in-suite
Theme injection into proxied WebUI (plugins/theme_injector.py)	implemented yes	integration test_bridge_theme_injection.py	tested-green-in-suite

2. qBitTorrent-go backend (Go/Gin) â€” opt-in --profile go

Entry: qBitTorrent-go/cmd/qbittorrent-proxy/main.go (routes registered :54-101). **Skeleton/rewrite-in-progress** (per AGENTS.md + docs/migration/PARITY_GAPS.md). NOT running by default.

2a. HTTP handlers (internal/api/*.go)

Feature	Impl	Wired Tests	Validation
GET /health â€” HealthHandler (api/health.go:5)	implemented opt-in	go internal/api/api_test.go	unit-only (go test)

Feature	Impl	Wired Tests	Validation
GET /api/v1/config â€” ConfigHandler (api/ download.go:210)	implemented	opt-in go coverage_test.go	unit-only
GET /api/v1/bridge/ health â€” BridgeHealthHandler (api/download.go:189)	implemented	opt-in go coverage_test.go	unit-only
POST /api/v1/search â€” SearchHandler (api/ search.go:14)	implemented (proxies qBit search; no plugin fan- out)	opt-in go api_test.go, coverage_test.go	unit-only
POST /api/v1/search/ sync â€” SearchSyncHandler (api/search.go:50)	implemented	opt-in go coverage_test.go	unit-only
GET /api/v1/search/ stream/:id â€” SearchStreamHandler SSE (api/search.go:91)	implemented	opt-in go coverage_test.go	unit-only
GET /api/v1/search/:id â€” GetSearchHandler (api/search.go:144)	implemented	opt-in go getsearch_results_test.go, coverage_test.go	unit-only
POST /api/v1/ search/:id/abort â€” AbortSearchHandler (api/search.go:177)	implemented	opt-in go coverage_test.go	unit-only
POST /api/v1/download â€” DownloadHandler (api/download.go:17)	partial â€” returns mock success, no	opt-in go coverage_test.go	unit-only function hole

Feature	Impl	Wired Tests	Validation
	qBit integration		
POST /api/v1/download/ file â€” DownloadFileHandler (api/download.go:41)	implemented	opt-in go coverage_test.go	unit-onl
POST /api/v1/magnet â€” MagnetHandler (api/ download.go:86)	implemented	opt-in go api_test.go, magnet_stress_chaos_test.go	unit + stress- chaos (
GET /api/v1/downloads/ active â€” ActiveDownloadsHandler (api/download.go:147)	stub â€” returns empty array, no qBit query	opt-in go coverage_test.go	unit-onl function hole
POST /api/v1/auth/ qbittorrent â€” QBittorrentAuthHandler (api/download.go:156)	implemented	opt-in go coverage_test.go	unit-onl
GET /api/v1/theme â€” GetThemeHandler (api/ theme.go:90)	implemented	opt-in go theme_hardening_test.go, coverage_test.go	unit-onl
PUT /api/v1/theme â€” PutThemeHandler (api/ theme.go:96)	implemented	opt-in go theme_hardening_test.go	unit-onl
GET /api/v1/hooks â€” ListHooksHandler (api/ hooks.go:91)	implemented (store only, no dispatch)	opt-in go api_test.go, coverage_test.go	unit-onl
	implemented (store only)	opt-in go api_test.go	unit-onl

Feature	Impl	Wired Tests	Validation
POST /api/v1/hooks "CreateHookHandler (api/hooks.go:97)			
DELETE /api/v1/hooks/:id "DeleteHookHandler (api/hooks.go:109)	implemented (store only)	opt-in go api_test.go	unit-only
GET /api/v1/schedules "ListSchedulesHandler (api/scheduler_api.go:86)	implemented (store only)	opt-in go coverage_test.go	unit-only no driver loop (RW-10) scheduler never f
POST /api/v1/schedules "CreateScheduleHandler (api/scheduler_api.go:92)	implemented (store only)	opt-in go coverage_test.go	unit-only never f (RW-10)
DELETE /api/v1/schedules/:id "DeleteScheduleHandler (api/scheduler_api.go:104)	implemented (store only)	opt-in go coverage_test.go	unit-only

2b. qBittorrent Web API client (internal/client/*.go)

Feature	Impl	Wired Tests	Validation	Video
NewClient "construct + login (client/client.go:18)	implemented	opt-in go client/client_test.go	unit-only	N/A (no UI "go-test-covered client method; opt-in profile)
Login " POST /api/v2/auth/login, store SID	implemented	opt-in go client/auth_test.go	unit-only	N/A (no UI "go-test-

Feature	Impl	Wired Tests	Validation	Video
(client/ auth.go:11)				covered client method; opt-in profile)
IsAuthenticated â€” SID presence (client/ auth.go:43)	implemented opt-in	go client/ auth_test.go	unit-only	N/A (no UI â€” go-test- covered client method; opt-in profile)
GetSID â€” thread-safe SID accessor (client/ client.go:39)	implemented opt-in	go client/ client_test.go	unit-only	N/A (no UI â€” go-test- covered client method; opt-in profile)
GetTorrents â€” GET /api/v2/ torrents/info (client/ torrents.go:13)	implemented opt-in	go client/ coverage_test.go	unit-only	N/A (no UI â€” go-test- covered client method; opt-in profile)
AddTorrent â€” POST /api/v2/ torrents/add URL (client/ torrents.go:30)	implemented opt-in	go client/ coverage_test.go	unit-only	N/A (no UI â€” go-test- covered client method; opt-in profile)
AddTorrentFile â€” POST multipart /api/ v2/torrents/add (client/ torrents.go:54)	implemented opt-in	go client/ coverage_test.go	unit-only	N/A (no UI â€” go-test- covered client method; opt-in profile)
GetAppVersion â€” GET /api/v2/ app/version	implemented opt-in	go client/ coverage_test.go	unit-only	N/A (no UI â€” go-test- covered

Feature	Impl	Wired Tests	Validation	Video
(client/ torrents.go:87)				client method; opt-in profile)
StartSearch â€” POST /api/v2/ search/start (client/ search.go:27)	implemented	opt-in go client/ coverage_test.go	unit-only	N/A (no UI â€” go-test- covered client method; opt-in profile)
GetSearchResults â€” GET /api/v2/ search/results (client/ search.go:55)	implemented	opt-in go client/ coverage_test.go	unit-only	N/A (no UI â€” go-test- covered client method; opt-in profile)
StopSearch â€” POST /api/v2/ search/stop (client/ search.go:84)	implemented	opt-in go client/ coverage_test.go	unit-only	N/A (no UI â€” go-test- covered client method; opt-in profile)
SearchStatus â€” GET /api/v2/ search/status (client/ search.go:100)	implemented	opt-in go client/ coverage_test.go	unit-only	N/A (no UI â€” go-test- covered client method; opt-in profile)
ListPlugins â€” GET /api/v2/ search/plugins (client/ search.go:121)	implemented	opt-in go client/ coverage_test.go	unit-only	N/A (no UI â€” go-test- covered client method; opt-in profile)

2c. Services (internal/service/*.go)

Feature	Impl	Wired Tests
		opt-in

Feature	Impl	Wired Tests	
Merge search orchestrator RunSearch poll/dedup loop (service/ merge_search.go:183)	implemented (qBit search only; no python-style plugin fan-out)	go service/service_test.go, runsearch_dedup_test.go	
StartSearch / GetSearchStatus / AbortSearch lifecycle (service/ merge_search.go:101-143)	implemented	opt-in	go getsearchstatus_hardening_t
Dedup key (FileURL+name+size) (service/ merge_search.go:269)	implemented	opt-in	go runsearch_dedup_test.go
FetchTorrent tracker fetch (service/ merge_search.go:276)	stub “ returns “not yet implemented“	opt-in	none
Stats counters (service/ merge_search.go:280)	implemented	opt-in	go coverage_test.go
SSE broker pub/sub (service/sse_broker.go)	implemented (defined but not wired “ handlers SSE inline)	opt-in	none dedicated
Metadata enricher (Go)	missing (RW-11)	no	none
Scheduler driver loop (Go)	missing (RW-10) “ CRUD only, no cron execution	no	none

2d. webui-bridge binary (Go) + cross-cutting

Feature	Impl	Wired Tests		Validation	Video
webui-bridge binary (cmd/ webui- bridge/ main.go)	partial (health + root only)	opt-in	none dedicated	not- validated	N/A (no UI of its own “ proxy binary; opt- in profile)
CORS + logging middleware	implemented	opt-in	go middleware/ middleware_test.go	unit-only	N/A (no UI “ go-test- covered middleware;

Feature	Impl	Wired Tests	Validation	Video
(internal/ middleware/)				opt-in profile)
Models / data types (internal/ models/)	implemented	opt-in go models/ models_test.go	unit-only	N/A (no UI â€” go-test- covered data types; opt-in profile)
Env config loader (internal/ config/)	implemented	opt-in go config/ config_test.go	unit-only	N/A (no UI â€” go-test- covered config loader; opt- in profile)
Log redactor (internal/ logging/ redactor.go)	implemented	opt-in go logging/ redactor_test.go	unit-only	N/A (no UI â€” go-test- covered redactor; opt-in profile)

3. boba-jackett (Go) â€” :7189

Entry: qBittorrent-go/cmd/boba-jackett/main.go; router internal/
jackettapi/router.go. Owns Jackett credentials + indexer overrides +
autoconfig run history; encrypted SQLite at /config/boba.db.

3a. HTTP endpoints (internal/jackettapi/router.go + handlers)

Feature	Impl	Wired Tests	Validation	V
GET /healthz (router.go:60, health.go)	implemented	yes go jackettapi/ health_test.go	unit (go)	N â€” c e
GET /openapi.json (router.go:64, openapi.go)	implemented	yes go jackettapi/ openapi_test.go; contract tests/contract/ openapi_test.go	unit + contract (go)	N â€” + c e
GET /api/v1/ jackett/ credentials â€” list (router.go:67, credentials.go)	implemented	yes go credentials_test.go; integration tests/ integration/ jackett_db_test.go	unit + integration (go)	N â€” c e b j

Feature	Impl	Wired Tests	Validation V
POST /api/v1/ jackett/ credentials "create/update" (router.go:67, credentials.go)	implemented yes	go credentials_test.go	unit + integration (go)
DELETE /api/v1/ jackett/ credentials/{name} (router.go:77, credentials.go)	implemented yes	go credentials_test.go	unit + integration (go)
GET /api/v1/ jackett/indexers "list configured" (router.go:86, indexers.go)	implemented yes	go jackettapi/ indexers_test.go	unit (go)
POST/PATCH/ DELETE /api/v1/ jackett/indexers/ {id} (router.go:93, indexers.go)	implemented yes	go indexers_test.go	unit (go)
POST /api/v1/ jackett/indexers/ {id}/test "test config" (indexers.go:239)	implemented yes	go indexers_test.go	unit (go)
GET /api/v1/ jackett/catalog "indexer catalog" (router.go:112, catalog.go)	implemented yes	go jackettapi/ catalog_test.go; benchmark catalog_bench_test.go	unit + bench (go)

Feature	Impl	Wired Tests	Validation V
POST /api/v1/jackett/catalog/refresh (router.go:119, catalog.go)	implemented yes	go catalog_test.go	unit (go)
GET /api/v1/jackett/autoconfig/runs "run history" (router.go:137, runs.go)	implemented yes	go jackettapi/runs_test.go; e2e tests/e2e/jackett_management_test.go	unit + e2e (go)
GET /api/v1/jackett/autoconfig/runs/{id} "run detail" (router.go:144, runs.go)	implemented yes	go runs_test.go	unit (go)
POST /api/v1/jackett/autoconfig/run "trigger run" (router.go:151, runs.go)	implemented yes	go runs_test.go; e2e tests/e2e/jackett_management_test.go	unit + e2e (go)
GET/POST /api/v1/jackett/overrides (router.go:160, overrides.go)	implemented yes	go jackettapi/overrides_test.go	unit (go)
DELETE /api/v1/jackett/overrides/{env} (router.go:170, overrides.go)	implemented yes	go overrides_test.go	unit (go)
Admin auth middleware (admin/admin on mutating routes) (jackettapi/auth_middleware.go)	implemented yes	go auth_middleware_test.go	unit (go)
Hardened CORS middleware (jackettapi/cors_middleware.go)	implemented yes	go cors_middleware_test.go	unit (go)

3b. Autoconfig engine + persistence (internal/jackett/, internal/db/)

Feature	Impl	Wired Tests	Validation	Video
Jackett autoconfig engine (jackett/ autoconfig.go)	implemented yes	go jackett/ autoconfig_test.go, autoconfig_bench_test.go	unit + bench (go)	N/A (no Uâ€” go-test-covered engine; exercised by web / jackett autoconfi run)
Fuzzy indexer matcher (jackett/ matcher.go)	implemented yes	go jackett/ matcher_test.go	unit (go)	N/A (no Uâ€” go-test-covered matcher)
Jackett HTTP client (jackett/ client.go)	implemented yes	go jackett/ client_test.go	unit (go)	N/A (no Uâ€” go-test-covered client)
Encrypted SQLite â€” AES-256-GCM crypto (db/ crypto.go)	implemented yes	go db/crypto_test.go; security tests/security/ credential_leak_test.go	unit + security (go)	N/A (no Uâ€” go-te + securit covered crypto)
SQLite migrations (db/ migrate.go)	implemented yes	go db/migrate_test.go	unit (go)	N/A (no Uâ€” go-test-covered migration)
Credential / indexer / runs repos (db/ repos/*.go)	implemented yes	go db/repos/*_test.go	unit (go)	N/A (no Uâ€” go-test-covered repos)
.env bootstrap import (bootstrap/ *.go)	implemented yes	go bootstrap/ bootstrap_test.go	unit (go)	N/A (no Uâ€” go-test-covered bootstrap)
Master-key ensure (first-boot) (bootstrap/ *.go)	implemented yes	go bootstrap/ bootstrap_test.go	unit (go)	N/A (no Uâ€” go-test-covered first-boot)

Feature	Impl	Wired Tests	Validation	Video
.env file parse/ write (envfile/ *.go)	implemented yes	go envfile/ parse_test.go, write_test.go	unit (go)	key ensure) N/A (no U â€” go- test- covered env file I/ O)

4. Tracker plugins (plugins/*.py)

Plugin contract: class with url, name, supported_categories, search(), download_torrent() â€” the *target* contract, not a description of every engine today (see CLAUDE.md Â§ Plugin System for the measured exceptions). Installed into the container by install-plugin.sh. Three counts, each answering a different question:

- **43 search-plugin engines** named by the PLUGINS=() array â€” the canonical managed roster.
- **43 distinct engine modules** on disk across plugins/, plugins/community/ and plugins/webui_compatible/; measured 2026-08-21, this is exactly the curated array.
- **36 *.py files** sit directly in plugins/ â€” the engines plus support modules (nova2, novaprinter, socks, helpers, env_loader, theme_injector, download_proxy), which are NOT plugins.

Corrected per BOB-149: the previous revision stated 44 and 27, said â€œ3 supportâ€” while listing 7, and claimed â€œmany curated names have NO matching file in this repo treeâ€”. That last claim was FALSE â€” every one of the 43 curated names resolves to a file in this tree (verified by set comparison, control-needle proven Â§11.4.201(7)(b)). These counts are guarded by scripts/pre_build/check_cm_plugin_count.sh.

4a. Plugins present in plugins/ AND named in the curated array

Plugin	Type	Impl	Wired (curated?)	Tests
eztv.py	public	implemented	yes (curated)	stress test_plugin_parsers
limetorrents.py	public	implemented		

Plugin	Type	Impl	Wired (curated?)	Tests
			yes (curated)	stress test_plugin_parsers word tests/unit/ test_plugin_multiword_qu stress/chaos tests/stress_ test_multiword_query_enc (Â§11.4.85)
piratebay.py	public	implemented	yes (curated)	stress test_plugin_parsers
solidtorrents.py	public	implemented	yes (curated)	stress test_plugin_parsers
torlock.py	public	implemented	yes (curated)	stress test_plugin_parsers
rutor.py	public (anonymous)	implemented	yes (curated)	stress test_plugin_parsers
nyaa.py	public	implemented	yes (curated)	stress test_plugin_parsers

Plugin	Type	Impl	Wired (curated?)	Tests
anilibra.py	public	implemented	yes (curated)	stress test_plugin_parsers
bitsearch.py	public	implemented	yes (curated)	stress test_plugin_parsers
gamestorrents.py	public	implemented	yes (curated)	stress test_plugin_parsers
kickass.py	public	implemented (degrades to honest empty)	yes (curated)	stress test_plugin_parsers

Plugin	Type	Impl	Wired (curated?)	Tests
megapeer.py	public	implemented	yes (curated)	stress test_plugin_parsers
tokyotoshokan.py	public	implemented	yes (curated)	stress test_plugin_parsers
torrentgalaxy.py	public	implemented	yes (curated)	stress test_plugin_parsers
torrentkitty.py	public	implemented	yes (curated)	stress test_plugin_parsers
yts.py	public	implemented	yes (curated)	stress test_plugin_parsers
jackett.py	aggregator	implemented	yes (curated)	covered via jackett autoconfi

Plugin	Type	Impl	Wired (curated?)	Tests
iptorrents.py	private (freeleech-only)	implemented	yes (curated)	integration tests/integrat
rutracker.py	private (cookies / CAPTCHA)	implemented	yes (curated)	integration test_tracker_a injection tests/unit/ rutrac stress test_plugin_parsers (ReDoS Â£11.4.85); ReDoS-d challenges/scripts/ rutracker_redos_regex_bo
kinozal.py	private	implemented	yes (curated)	integration test_tracker_a
nnmclub.py	private (cookies)	implemented	yes (curated)	integration test_tracker_a

Plugin	Type	Impl	Wired (curated?) Tests
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4b. Plugin support modules

Module	Type	Impl	Wired Tests	Validation	Validation
nova2.py " nova3 search harness	support implemented	yes	covered via search e2e	tested-green-in-suite	N "â€ co se ha ex by se flo N "â€ co re en ex by se flo N "â€ co he ex by se flo N "â€ in S su co in N "â€ co cr
novaprinter.py " result emitter (NDJSON-patched)	support implemented	yes	covered via search e2e	tested-green-in-suite	
helpers.py " HTTP/retrieve helpers	support implemented	yes	covered via search e2e	tested-green-in-suite	
socks.py " SOCKS proxy support	support implemented	yes	covered indirectly	not-validated	
env_loader.py " env/credential loader for plugins	support implemented	yes	covered via tracker auth tests	tested-green-in-suite	

Module	Type	Impl	Wired Tests	Validation	Video
Per-plugin download_torrent() path (.torrent fetch)	feature	implemented	yes integration test_buttons_api.py (download/file)	tested- green-in- suite	lo ex by Th ta N â€ co ex by re D bu N â€ co ex by re D (f po
Freeleech-only download policy (IPTorrents [free] tag)	policy	implemented	yes integration test_iporrents.py	tested- green-in- suite	

4c. Discrepancy rows (honest, Â§11.4.6)

Plugin	Type	Impl	Wired (curated?)	Tests	Validation	Video
torrentproject / torrentscsv	public (curated names)	named in install- plugin.sh PLUGINS array but NO matching plugins/ *.py file in tree	curated- name-only	none	not- implemented- in-repo-tree (install script expects them elsewhere)	N/A (not implemented in repo tree â€” curated name, no plugin file)
23 other curated names absent from tree: academictorrents, ali213, audiobookbay, bitru, bt4g, btsow, extratorrent, glotorrents, linuxtracker, one337x,	public (curated names)	named in curated array; NO plugins/ *.py in this tree	curated- name-only	none	not- implemented- in-repo-tree (name in install file absent)	N/A (not implemented in repo tree â€” curated names, no plugin files)

Plugin	Type	Impl	Wired (curated?)	Tests Validation	Video
pctorrent, pirateiro, rockbox, snowfl, therarbg, torrentdownload, torrentfunk, xfsb, yihua, yourbittorrent (+ torrentproject, torrentscsv)					

NOTE (Â§11.4.6, CORRECTION vs Rev 2): the install-plugin.sh PLUGINS array contains **44 entries**, not 12. Of those, ~21 have a matching plugins/*.py file in this repo; the rest are curated names with no file present in the tree. Stated as fact, not asserted as working.

5. Angular 21 frontend dashboard (frontend/)

Standalone SPA, signals-based; built to download-proxy/src/ui/dist/frontend, served from :7187. Vitest unit tests; Playwright e2e under frontend/e2e.

5a. Dashboard component controls (components/dashboard/dashboard.component.ts)

Feature	Impl	Wired Tests
Search box (query signal) + submit	implemented yes	vitest dashboard.component.spec.ts; py integration test_dashboard_rendering.py, test_ui_comprehensive.py
Result grid “sortable (name/type/size/seeds/leechers/quality/sources)	implemented yes	vitest dashboard.component.spec.ts

Feature	Impl	Wired Tests
Results tab	implemented yes	py test_ui_comprehensive.py
Active Downloads tab + view (dashboard.component.ts:745 loadDownloads)	implemented yes	py test_buttons_api.py
Trackers tab + per-tracker auth chips (expandable)	implemented yes	py test_auth_state_ui.py
Schedules tab	implemented yes	py test_dashboard_automation.py
Hooks tab	implemented yes	py test_dashboard_automation.py
Per-row Magnet button â†’ magnet dialog trigger	implemented yes	vitest magnet- dialog.component.spec.ts; py test_magnet_dialog.py
Per-row add-to-qBittorrent (qBit) button	implemented yes	py test_button_functions.py, test_buttons_api.py
Per-row Download (.torrent) button	implemented yes	py test_buttons_api.py

Feature	Impl	Wired Tests
Per-row processing spinner (isBusy())	implemented yes	vitest dashboard.component.spec.ts
Bridge health indicator + retry button	implemented yes	py test_bridge_root_liveness.py
Tracker-stat dialog trigger	implemented yes	py contract test_tracker_stats_contract.p

5b. Dialog + UI components

Feature	Impl	Wired Tests	Val
Magnet dialog “ readonly textarea + Copy/ Open/Add/Close (magnet- dialog.component.ts)	implemented yes	vitest magnet- dialog.component.spec.ts; py test_magnet_dialog.py	tes gre sui
qBit login dialog “ user/ pass/remember/Login/ Cancel (qbit-login- dialog.component.ts)	implemented yes	vitest qbit-login- dialog.component.spec.ts; py test_login_actions.py, test_auth_state_ui.py	tes gre sui

Feature	Impl	Wired Tests	Validation
Theme picker â€” palette dropdown + light/dark toggle (theme-picker.component.ts)	implemented yes	vitest theme-picker.component.spec.ts; py e2e test_theme_runtime.py, test_crossapp_theme.py	tested-green-suite
Confirm dialog â€” title/message/Confirm/Cancel/backdrop (confirm-dialog.component.ts)	implemented yes	vitest confirm-dialog.component.spec.ts	unit (vitest)
Tracker-stat dialog â€” JSON view + Copy + status badge + notes table + Esc-close (tracker-stat-dialog.component.ts)	implemented yes	vitest tracker-stat-dialog.component.spec.ts; py contract test_tracker_stats_contract.py	tested-green-suite
Toast container â€” info/success/warning/error auto-dismiss (toast-container.component.ts)	implemented yes	vitest toast-container.component.spec.ts	unit (vitest)
Site footer (site-footer.component.ts)	implemented yes	vitest	unit (vitest)

5c. Services (frontend/src/app/services/)

Feature	Impl	Wired Tests	Validation
API service â€” HTTP wrapper (search/download/magnet/auth/schedules/hooks/downloads/stats/config), 15s timeout, runtime baseUrl (api.service.ts)	implemented yes	vitest api.service.spec.ts	unit (vitest)
SSE service â€” EventSource live result streaming + bearer-token (sse.service.ts)	implemented yes	vitest sse.service.spec.ts; py integration test_realtime_streaming.py, test_streaming_browser.py	tested-green-in-suite
Theme service â€” palette/mode + localStorage + cross-	implemented yes	vitest theme.service.spec.ts; py	tested-green-in-suite

Feature	Impl	Wired Tests	Validation V
app sync (theme.service.ts)		e2e test_theme_runtime.py, test_crossapp_theme.py	
Toast service â€” toast queue state (toast.service.ts)	implemented yes	vitest toast.service.spec.ts	unit (vitest)
Dialog service â€” promise-based confirm state (dialog.service.ts)	implemented yes	vitest dialog.service.spec.ts	unit (vitest)

5d. Jackett page (frontend/src/app/jackett/)

Feature	Impl	Wired Tests	Va
Credentials â€” list + Add/ Edit/Delete (per-row, window.confirm) + banners (jackett/credentials/ credentials.component.ts)	implemented yes	vitest credentials.component.spec.ts	un (v
Credential edit dialog â€” name/username/password/ cookies/Save/Cancel (credential-edit- dialog.component.ts)	implemented yes	vitest credential-edit- dialog.component.spec.ts	un (v
Indexers â€” Configured/ Catalog/History tab container (jackett/ indexers/ indexers.component.ts)	implemented yes	vitest indexers.component.spec.ts	un (v
Indexers â€” Configured tab (list + per-row remove)	implemented yes	vitest indexers.component.spec.ts	un (v

Feature	Impl	Wired Tests	Validation
Indexers â€” Catalog tab (list + per-row add)	implemented yes	vitest indexers.component.spec.ts	untested (v8)
Indexers â€” History tab (autoconfig run history)	implemented yes	vitest indexers.component.spec.ts	untested (v8)
Indexer add dialog â€” credential dropdown + Save/Cancel + error banner (indexer-add-dialog.component.ts)	implemented yes	vitest indexer-add-dialog.component.spec.ts	untested (v8)
IPTorrents cookie-flow component (iptorrents-cookie-flow.component.ts)	implemented yes	vitest (jackett specs)	untested (v8)

5e. Frontend infra

Feature	Impl	Wired	Tests	Validation	Video
Frontend coverage thresholds (40% lines/branches/funcs/stmts, v8)	infra	yes	vitest.config.ts	enforced in vitest config	N/A (no UI â€” build/coverage infra)

6. BobaLink browser extension (extension/)

WXT + TypeScript Manifest-V3. Detects magnet/.torrent links and forwards to merge service :7187. Status detail: docs/browser_extension/Status.md. Built zips at extension/.output/bobalink-1.0.0-{chrome,firefox}.zip.

6a. User-facing surfaces

Feature	Impl	Wired Tests	Validation	Video
Popup â€” per-row Send button (src/popup/entrypoints/popup.ts)	implemented yes	a11y tests/a11y/popup.a11y.test.ts	tested-green-in-suite	VIDEO-CONFERENCE 2026-06-16 extension-p verdict docs/recordings-boba-extension-verdict.md
Popup â€” Send-All button	implemented yes	a11y popup.a11y.test.ts	tested-green-in-suite	VIDEO-CONFERENCE 2026-06-16 extension-p verdict docs/recordings-boba-extension-verdict.md

Feature	Impl	Wired Tests	Validation	Video
Popup "Refresh button + status indicator (online/offline/warning)	implemented yes	a11y popup.a11y.test.ts	tested-green-in-suite	VIDEO-CONFERENCE 2026-06-16 extension-podcasts-verdict/docs/recordings-boba-extension-verdict.md
Popup "torrent list (name + type + infohash prefix + sent badge)	implemented yes	a11y popup.a11y.test.ts	tested-green-in-suite	VIDEO-CONFERENCE 2026-06-16 extension-podcasts-verdict/docs/recordings-boba-extension-verdict.md
Popup "Open Options link	implemented yes	a11y popup.a11y.test.ts	tested-green-in-suite	VIDEO-CONFERENCE 2026-06-16 extension-podcasts-verdict/docs/recordings-boba-extension-verdict.md
Options page (src/options/options.ts)	implemented yes	a11y options.a11y.test.ts; integration options-background.test.ts	tested-green-in-suite	VIDEO-CONFERENCE 2026-06-16 extension-podcasts-options.mp4/docs/qa/recordings-boba-extension-verdict.md
Content script "link/text highlight overlay (src/content/highlight.ts, entrypoints/content.ts)	implemented yes	integration content-background.test.ts; security content-xss.test.ts	tested-green-in-suite	VIDEO-CONFERENCE 2026-06-16 extension-podcasts-detect.mp4/docs/qa/recordings-boba-extension-verdict.md
Background service worker "message hub (src/background/, entrypoints/background.ts)	implemented yes	integration tests/integration/*-background.test.ts	tested-green-in-suite	N/A (no UI "covered + extension scalability)
Context-menu entries (right-click torrent actions)	implemented yes implemented yes	integration *-background.test.ts	tested-green-in-suite	N/A (no UI "covered + extension scalability)

Feature	Impl	Wired Tests	Validation Video	
Keyboard shortcuts (scan-now / highlight-toggle / send-all)	implemented yes	integration *-background.test.ts	tested-green-in-suite	N/A (no UI â€” covered + extension sca
Action badge (detected count) + notifications		integration *-background.test.ts	tested-green-in-suite	N/A (no UI â€” covered + extension sca

6b. Scanners (src/scanner/)

Feature	Impl	Wired Tests	Validation Video	
Base scanner â€” computeStableId() lifecycle (scanner/base.ts)	implemented yes	vitest unit	tested-green-in-suite	N/A (no UI â€” test-covered scanner; exercised by extension scan-detect)
Link scanner â€” <a href>magnet/.torrent (scanner/link-scanner.ts)	implemented yes	security scanner-hostile-input.test.ts; perf scanner.perf.test.ts	tested-green-in-suite	N/A (no UI â€” test-covered scanner; exercised by extension scan-detect)
Text scanner â€” bare magnet text (scanner/text-scanner.ts)	implemented yes	security scanner-hostile-input.test.ts	tested-green-in-suite	N/A (no UI â€” test-covered scanner; exercised by extension scan-detect)
Scanner orchestrator â€” dedup + mutation-observer re-scan (scanner/orchestrator.ts)	implemented yes	perf orchestrator-scaling.perf.test.ts	tested-green-in-suite	N/A (no UI â€” test-covered scanner; exercised

Feature	Impl	Wired Tests	Validation	Video
Site DB “ per-site selector match rules (scanner/site-db.ts)	implemented	yes vitest unit	tested-green-in-suite	by extension scan-detect) N/A (no UI “ test-covered scanner; exercised by extension scan-detect)

6c. Parsers (src/parser/)

Feature	Impl	Wired Tests	Validation	Video
Bencode parser (parser/bencode.ts)	implemented	yes security bencode-torrentfile-hostile.test.ts; perf parsers.perf.test.ts	tested-green-in-suite	N/A (no UI “ test-covered parser; exercised by extension scan-detect)
Magnet parser (parser/magnet.ts)	implemented	yes security infohash-detection-hostile.test.ts; perf magnet.perf.test.ts	tested-green-in-suite	N/A (no UI “ test-covered parser; exercised by extension scan-detect)
Torrent-file parser (parser/torrent-file.ts)	implemented	yes security bencode-torrentfile-hostile.test.ts	tested-green-in-suite	N/A (no UI “ test-covered parser; exercised by extension scan-detect)

6d. API client + offline (src/api/)

Feature	Impl	Wired Tests	Validation	Video
Boba API client â€” POST /api/v1/download, retry + rate-limit (api/ boba-client.ts)	implemented	yes vitest unit; chaos tests/ chaos/boba-client- resilience.chaos.test.ts	tested- green-in- suite	N/A (no UI â€” test- covered API client; exercised by extension Send)
Health probe â€” probeHealth() (api/ health.ts)	implemented	yes vitest unit	tested- green-in- suite	N/A (no UI â€” test- covered health probe; exercised by extension status indicator)
Offline send queue â€” IndexedDB enqueue + re- dispatch (api/ queue.ts)	implemented	yes chaos tests/chaos/ queue.chaos.test.ts	tested- green-in- suite	N/A (no UI â€” test- covered offline queue; chaos- covered)

6e. Shared utilities + tabgroups (src/shared/, src/ tabgroups/)

Feature	Impl	Wired Tests	Validation	Video
Crypto â€” AES-256- GCM token encrypt/ decrypt (shared/ crypto.ts)	implemented	yes security crypto- tamper.test.ts, no-hardcoded- secret.test.ts	tested- green-in- suite	N/A (no UI â€” test- covered crypto; exercised by extension Send token path)
Storage â€” namespaced	implemented	yes vitest unit		N/A (no UI â€” test- covered)

Feature	Impl	Wired Tests	Validation	Video
chrome.storage.local wrapper (shared/storage.ts)			tested-green-in-suite	test-covered storage wrapper)
Constants â€” site selectors / rate-limit / retry config (shared/constants.ts)	implemented yes	covered via scanner tests	tested-green-in-suite	N/A (no UI â€” test-covered constants)
Errors â€” Network/Server/Storage error types (shared/errors.ts)	implemented yes	vitest unit	tested-green-in-suite	N/A (no UI â€” test-covered error types)
Events â€” TypedEventEmitter bus (shared/events.ts)	implemented yes	vitest unit	tested-green-in-suite	N/A (no UI â€” test-covered event bus)
Logger â€” namespaced, no-secret (shared/logger.ts)	implemented yes	vitest unit	tested-green-in-suite	N/A (no UI â€” test-covered logger)
Utils â€” debounce / TokenBucket rate limiter (shared/utils.ts)	implemented yes	vitest unit	tested-green-in-suite	N/A (no UI â€” test-covered utils/rate-limiter)
Tab groups â€” batch grouped-tab torrents (tabgroups/index.ts)	implemented yes	vitest unit (tab-group Challenge per ext Status)	tested-green-in-suite	N/A (no UI of its own â€” test-covered tab-group batching; exercised by extension Send-All)

6f. i18n locales (extension/src/public/_locales/)

Feature	Impl	Wired Tests	Validation	Video
	implemented yes			

Feature	Impl	Wired	Tests	Validation	Video
Locale en (English)			i18n tests/ i18n/locale- safety.test.ts	tested-green- in-suite	N/A (no UI of its own â€” i18n locale data; test- covered; surfaced inside extension popup/ options)
Locale de (German)	implemented	yes	i18n locale- safety.test.ts	tested-green- in-suite	N/A (no UI of its own â€” i18n locale data; test- covered; surfaced inside extension popup/ options)
Locale es (Spanish)	implemented	yes	i18n locale- safety.test.ts	tested-green- in-suite	N/A (no UI of its own â€” i18n locale data; test- covered; surfaced inside extension popup/ options)
Locale fr (French)	implemented	yes	i18n locale- safety.test.ts	tested-green- in-suite	N/A (no UI of its own â€” i18n locale data; test- covered; surfaced inside extension

Feature	Impl	Wired	Tests	Validation	Video
Locale it (Italian)	implemented	yes	i18n locale- safety.test.ts	tested-green- in-suite	popup/ options) N/A (no UI of its own "â€" i18n locale data; test- covered; surfaced inside extension popup/ options) N/A (no UI of its own "â€" i18n locale data; test- covered; surfaced inside extension popup/ options)
Locale ja (Japanese)	implemented	yes	i18n locale- safety.test.ts	tested-green- in-suite	popup/ options) N/A (no UI of its own "â€" i18n locale data; test- covered; surfaced inside extension popup/ options) N/A (no UI of its own "â€" i18n locale data; test- covered; surfaced inside extension popup/ options)
Locale pt (Portuguese)	implemented	yes	i18n locale- safety.test.ts	tested-green- in-suite	popup/ options) N/A (no UI of its own "â€" i18n locale data; test- covered; surfaced inside extension popup/ options) N/A (no UI of its own "â€" i18n locale data; test- covered; surfaced inside extension popup/ options)
Locale ru (Russian)	implemented	yes	i18n locale- safety.test.ts	tested-green- in-suite	popup/ options) N/A (no UI of its own "â€" i18n locale data; test- covered; surfaced

Feature	Impl	Wired Tests	Validation	Video inside extension popup/ options)
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6g. Live round-trip

Feature	Impl	Wired Tests	Validation	Video
Live download endpoint round-trip (tests/live/ download- endpoint.live.test.ts)	implemented	yes (live- gated)	live download- endpoint.live.test.ts	partial "operator- gated live env"

7. WebUI bridge (webui-bridge.py) " host process :7188

Bridges qBittorrent WebUI with private-tracker auth. NOT a container.

Feature	Impl	Wired Tests	Validation
WebUI proxy + header rewrite (webui-bridge.py)	implemented	host integration test_webui_bridge.py, test_bridge_header_rewrite.py; stress test_bridge_stress_chaos.py	tested- green-in- suite
Private-tracker auth bridging (rutracker/ kinozal/nnmclub/ iptorrents)	implemented	host integration	test_webui_bridge.py partial "live-gated"
Theme injection into bridged WebUI (plugins/ theme_injector.py)	implemented	host integration test_bridge_theme_injection.py	tested- green-in- suite

Feature	Impl	Wired Tests	Validation
Bridge root liveness (/)	implemented host	integration test_bridge_root_liveness.py	tested-green-in-suite

8. Infrastructure / CLI / shell scripts

8a. boba-ctl CLI orchestrator (Go: cmd/boba-ctl/main.go + wrapper scripts/boba-ctl.sh)

Feature	Impl	Wired Tests	Validation	Video
boba-ctl up start stack (cmd/boba-ctl/main.go:32)	implemented host	go cmd/boba-ctl/projectroot_test.go	tested-green-in-suite	N/A (CLI "g covered; exercise boba-ctl CLI d boba-cli-orchestrator demo.mp4)
boba-ctl down " stop stack (main.go:34)	implemented host	go cmd/boba-ctl/projectroot_test.go	tested-green-in-suite	N/A (CLI "g covered; exercise boba-ctl CLI d boba-cli-orchestrator demo.mp4)
boba-ctl status service status (main.go:36)	implemented host	go cmd/boba-ctl/projectroot_test.go	tested-green-in-suite + LIVE	VIDEO-CONF 2026-06-16 â cli-orchestrator demo.mp4; ver docs/qa/recordings-2 boba-cli-ver
boba-ctl health health probe (main.go:38, cmd/boba-ctl/health_test.go)	implemented host	go cmd/boba-ctl/health_test.go	tested-green-in-suite + LIVE	VIDEO-CONF 2026-06-16 â cli-orchestrator demo.mp4
boba-ctl list " list services (main.go:40)	implemented host implemented host	go cmd/boba-ctl/projectroot_test.go	tested-green-in-suite + LIVE	VIDEO-CONF 2026-06-16 â cli-orchestrator demo.mp4

Feature	Impl	Wired Tests	Validation	Video
boba-ctl.sh wrapper “ compose-compatible shim (up/down/ ps/config/pull + passthrough) (scripts/boba- ctl.sh)		covered via boba-ctl go tests	tested- green-in- suite	N/A (CLI shim covered; exer boba-ctl CLI d

8b. Lifecycle scripts (repo root)

Feature	Impl	Wired Tests	Validation
start.sh “ container bring- up + Jackett key extraction/ injection	implemented host	covered via e2e test_fixtures_bring_up_services.py	tested- green-in- suite
stop.sh “ stop/remove/ purge	implemented host	none dedicated	not- validated
setup.sh “ one-time setup	implemented host	none dedicated	not- validated
start-proxy.sh “ proxy container entry helper	implemented host	none dedicated	not- validated
install- plugin.sh “ copy curated plugins “ container engines	implemented host	covered indirectly (e2e search)	tested- green-in- suite
init-qbit- password.sh “ qBit admin password bootstrap	implemented host	none dedicated	not- validated
fix-qbit- password.sh “ qBit admin password repair	implemented host	none dedicated	not- validated

Feature	Impl	Wired Tests	Validation	Video
setup-webui-bridge-service.sh + webui-bridge.service â€” systemd unit for bridge	implemented	host none dedicated		not-validated

8c. Test / CI wrappers (repo root)

Feature	Impl	Wired Tests	Validation	Video
ci.sh â€” manual CI (syntax+unit+integration+e2e+health)	implemented	host	self (is the gate) tested-green-in-suite	N/A (host CI/test wrapper â€” no UI; is the gate)
run-all-tests.sh â€” full suite (hardcoded podman)	implemented	host	self	partial â€” fails on docker-only hosts (known gotcha) N/A (host CI/test wrapper â€” no UI; is the gate)
test.sh â€” quick validation wrapper	implemented	host	self	tested-green-in-suite N/A (host CI/test wrapper â€” no UI; is the gate)
test-all.sh â€” full validation wrapper	implemented	host	self	tested-green-in-suite N/A (host CI/test wrapper â€” no UI; is the gate)
test-full.sh â€” extended validation wrapper	implemented	host	self	tested-green-in-suite N/A (host CI/test wrapper â€” no UI; is

Feature	Impl	Wired Tests	Validation Video the gate)
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8d. scripts/ helpers

Feature	Impl	Wired Tests
scripts/add-submodules.sh " submodule bootstrap	implemented host	none dedicated
scripts/audit-plugins.sh " plugin discrepancy audit	implemented host	none dedicated
scripts/build-releases.sh " build extension/Go release artifacts	implemented host	none dedicated
scripts/ codegraph_validate.sh " CodeGraph index validation (11.4.78)	implemented host	self
scripts/workable-items-export.sh " docs regeneration wrapper (11.4.93/11.4.65; renamed 2026-08-15 BOB-104 from docs_chain.sh " that name was a misnomer)	implemented host	self
constitution/submodules/docs_chain/docs_chain " real 11.4.106 Docs Chain engine (Go binary, pinned to helixcode-v1.1.0)	implemented host	self
scripts/ensure-macos-tunnel.sh " 0.0.0.0 SSH tunnel	implemented host	none dedicated (log qa-results, tunnel-keepalive.log)
scripts/tunnel-keepalive.sh " tunnel self-heal	implemented host	none dedicated

Feature	Impl	Wired Tests
scripts/freeze-openapi.sh â€” freeze boba-jackett OpenAPI snapshot	implemented host	contract tests/contract/openapi_test.go
scripts/ generate_markdown_exports.sh â€” mdâ†”html/pdf export (Â§11.4.65)	implemented host	self
scripts/helixqa.sh â€” HelixQA harness driver	implemented host	self
scripts/opencode-helixqa.sh â€” OpenCode HelixQA driver	implemented host	self
scripts/install_git_hooks.sh â€” git hook installer (Â§11.4.75)	implemented host	self
scripts/ pre_build_verification.sh â€” pre-build gate sweep	implemented host	self (is the gate)
scripts/pre_code_review.sh â€” code-review gate (Â§11.4.125)	implemented host	self
scripts/ run_all_challenges.sh â€” challenge bank runner	implemented host	self
scripts/run-tests.sh â€” test runner helper	implemented host	self
	implemented host	none dedicated

Feature	Impl	Wired Tests
scripts/scan.sh " repo scan helper		
scripts/track-build-resources.sh " build resource sampler (11.4.24)	implemented host	none dedicated
scripts/load-tracker-cookies.sh " per-tracker Netscape cookies-file autoload " .env (operator mandate 2026-08-15, 619a5f6)	implemented host	challenge challenges/scripts/credentials_wired_challenge.
scripts/boba-svc.sh " sudo-free systemd-linger service wrapper (up/down/status/logs)	implemented host	challenge challenges/scripts/install_sh_idempotent_challenge_credentials_wired_challenge.
scripts/commit-push-all.sh " 11.4.234 dedicated commit/push entrypoint (hook-validation stage + always-unblocked push, 08e870e)	implemented host	none dedicated

8e. Infra / compose / startup

Feature	Impl	Wired Tests	Validation
docker-compose.yml " 2-container default + Go profile + boba-jackett service	infra	yes e2e test_live_containers.py	tested-green-in-suite
Jackett auto-configuration at startup (key extract " inject JACKETT_API_KEY)	implemented yes	integration test_jackett_autoconfig_real.py	tested-green-in-suite
BOBA_MASTER_KEY auto-generation (first-boot) for encrypted boba.db	implemented yes	go bootstrap/bootstrap_test.go	unit (go)

Components NOT inventoried / discrepancies (honest gaps, Â§11.4.6)

- **Rev 9 scope boundary (BOB-083, 2026-08-20):** this pass corrected 2 pre-existing rows (RW-06 rutracker ReDoS-deploy row Â§4b, RW-07 nnmclub-status row Â§1b) to PASS â€” LIVE with real evidence citations (1c0389a/BOB-093, 7baef2b/BOB-092), but did **NOT** re-audit the other ~290 pre-existing rows against the 161-commit delta since Rev 8 (2026-08-18) â€” same bounded-scope reasoning as Rev 8. Known-real, non-feature work that landed in that window and is intentionally NOT itemized as new rows here (process-safety/CI discipline, not a product feature): the BOB-126 killpg(-1, SIGKILL) forced-logout incident chain root-cause + fix (ad4b46a, new universal Â§11.4.263 anchor) and its surrounding gate/test hardening (BOB-127..135, ~150 commits). A future pass MAY re-run the full inventory to close this gap properly; until then this document is accurate for what it explicitly claims and honestly incomplete for what it does not.
- **Rev 8 scope boundary (BOB-075, 2026-08-18):** this pass added 3 new rows for scripts that shipped with no prior row (Â§8d) and fixed the missing Rev/date bump for the ea86ce1 content edit, but did **NOT** re-audit the other ~289 pre-existing rows against the current tree â€” that would be a full re-run of the original READ-ONLY repo-inventory process (Â§11.4.118 discovery-pressure), out of scope for a bounded staleness fix. Known-real capabilities that shipped since Rev 7 and are **not yet itemized at row-granularity** here (tracked as an honest gap, not silently assumed covered): the systemd service topology (boba.target, boba-stack.service, boba-webui-bridge.service, 444d94e/bbc63ca), outbound egress-proxy routing for tracker fetches (BOBA_UPSTREAM_PROXY, b44a2e4/4e62eca/ be5062d), and the submodules/jackett version bump to v0.24.2406 (99a486e, BOB-076 â€” a dependency-version change, not a new feature row). A future pass MAY re-run the full inventory; until then this document is accurate for what it explicitly claims and honestly incomplete for what it does not.
- **CORRECTION vs Rev 2:** boba-ctl.sh DOES exist at scripts/boba-ctl.sh (Rev 2 said it did not). Its subcommands are now itemized in Â§8a. The underlying Go binary is cmd/boba-ctl/main.go (5 subcommands: up/down/status/health/list).
- **CORRECTION vs Rev 2:** the install-plugin.sh PLUGINS curated array has **44 entries**, not 12. ~21 have a matching plugins/*.py file; the rest (incl.Â torrentproject, torrentscsv) are curated names with no file present in this tree â€” itemized in Â§4c, not asserted as working.
- **Prometheus/OpenTelemetry metrics** â€” tests/observability/test_metrics_exist.py + observability/ dir exist, but no prometheus_client Counter()/Histogram()/Gauge() definitions were found in download-proxy/src; metrics surface is via GET /api/v1/stats (counters maintained in-process), not a /metrics Prometheus endpoint. Stated as fact, not assumed.

- **Go DownloadHandler / ActiveDownloadsHandler** â€” flagged partial/stub in Â§2a (mock success / empty array, no qBit integration) per source reading. The Go profile is opt-in and NOT the shipped product.
- **Go scheduler** â€” CRUD-only, no driver loop (RW-10); schedules never fire. The Go enricher is missing (RW-11).
- **Video-recording confirmation** â€” PENDING for all rows EXCEPT the boba-ctl CLI subcommands (status/health/list) and the web-UI flows already filmed (POST /api/v1/search journey + dashboard tab/button/theme/Jackett tour). Remaining per-feature flows are an in-progress recording pass (Â§11.4.107/Â§11.4.143).